

A negative literature answer to the tracial Markov-map question in arXiv:0803.2979

Literature-status note for `fa_banach_001`

11 August 2026

Status. `literature_already_answered` (negative).

Original question

After constructing factorizations for real positive Schur multipliers, Ricard observes that compositions in different bases, conditional expectations, representations, and convex combinations yield a wide class of factorizable maps. He then asks on page 4 of the arXiv PDF [1] whether these constructions can “achieve all tracial markovian maps for M_n .”

Every map produced in this way is factorizable: this is the point of the preceding construction and its permanence properties. Consequently, any tracial Markov map that is not factorizable is a counterexample to the question.

Explicit negative answer

Haagerup and Musat give exactly such a map in Example 3.1 on pages 9–10 of arXiv:1009.0778 [2]. Put

$$a_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad a_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad a_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and define

$$T(x) = \sum_{i=1}^3 a_i^* x a_i, \quad x \in M_3(\mathbb{C}).$$

Since

$$\sum_i a_i^* a_i = \sum_i a_i a_i^* = I_3,$$

T is completely positive, unital, and trace preserving, hence is a τ_3 -Markov map.

The same example proves that T is not factorizable. Indeed, their factorization criterion would give elements v_1, v_2, v_3 in a finite von Neumann algebra such that $u = \sum_i a_i \otimes v_i$ is unitary. The diagonal relations in $u^* u = I$ force $v_i^* v_i = I$, hence finiteness makes every v_i unitary. The off-diagonal relations simultaneously force

$$v_1^* v_2 = v_2^* v_3 = v_3^* v_1 = 0,$$

an impossibility for unitaries. Thus T is a tracial Markov map on M_3 outside the factorizable class, and so it cannot arise from Ricard’s listed construction. This answers the question negatively.

Dimension scope and identification

The introduction of the answering paper explicitly cites Ricard’s paper as the source for factorizability of real positive Schur multipliers and then announces nonfactorizable τ_n -Markov maps for every $n \geq 3$. It also records Kümmerer’s positive result that every trace-preserving Markov map on $M_2(\mathbb{C})$ is factorizable. Hence the later literature supplies the sharp dimension boundary: the exhaustion statement holds at $n = 2$ but fails for every $n \geq 3$.

The original question was checked on source PDF page 4. The counterexample and its contradiction proof were checked on supporting PDF pages 9–10; the all- $n \geq 3$ statement is in the abstract and Remark 4.8. This packet makes no new mathematical claim.

References

- [1] É. Ricard, *A Markov dilation for self-adjoint Schur multipliers*, Proc. Amer. Math. Soc. **136** (2008), 4365–4372; arXiv:0803.2979, <https://arxiv.org/abs/0803.2979>.
- [2] U. Haagerup and M. Musat, *Factorization and dilation problems for completely positive maps on von Neumann algebras*, Comm. Math. Phys. **303** (2011), 555–594; arXiv:1009.0778, <https://arxiv.org/abs/1009.0778>.